MINI PROJECT

Smart Bank Locker Access System with Password and RFID Authentication using LPC2129

Abstract

This project uses an LPC2129 (ARM7) microcontroller with an RFID reader and a keypad to build a two-level security system for a bank locker. The user must first scan an authorized RFID tag and then enter the correct password on the keypad. If both are correct, the DC motor unlocks the locker and a green LED glows. After three wrong password attempts, a buzzer sounds and the system locks itself. The LCD shows the access status, and every attempt is displayed on a PC through UART. The system gives a simple, low-cost and secure solution for lockers and safes.

Block Diagram

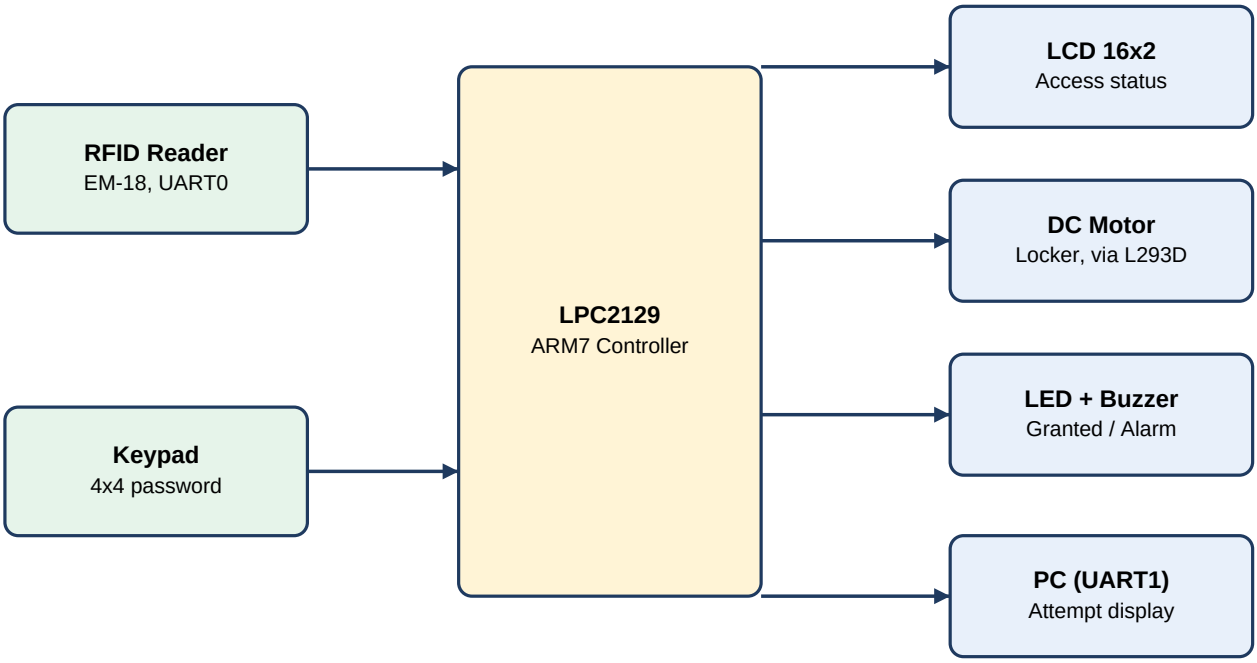

Fig. 1: Block diagram of the Smart Bank Locker Access System

Hardware Components

S.No	Component	Purpose
1	LPC2129 (ARM7)	Main microcontroller that controls authentication and all peripherals
2	EM-18 RFID Reader	Reads the RFID tag and sends its ID to the controller through UART0
3	4x4 Keypad	Accepts the password entered by the user
4	16x2 LCD	Displays the access status and messages to the user
5	DC Motor	Operates the locker mechanism to lock and unlock it
6	L293D Motor Driver	Drives the DC motor from the controller's output signals
7	LED	Green LED glows when access is granted
8	Buzzer	Sounds an alarm after three wrong password attempts
9	PC (UART1)	Displays every access attempt through serial communication

Description

The Smart Bank Locker Access System provides two levels of security before the locker can be opened. The LPC2129 ARM7 microcontroller acts as the central controller and coordinates all input and output devices.

Working:

1. **Level 1 - RFID authentication:** The user places an RFID tag near the EM-18 reader. The reader sends the tag ID to the LPC2129 through UART0, which checks it against the authorized tag.
2. **Level 2 - Password authentication:** If the tag is authorized, the user enters the password on the 4x4 keypad.
3. **Access granted:** If both the tag and the password are correct, the controller drives the DC motor through the L293D driver to unlock the locker. The green LED glows and the LCD shows the access status.
4. **Wrong password:** If the password is incorrect, the user can try again. After three wrong attempts, the buzzer sounds and the system locks itself.
5. **Monitoring:** Every access attempt is shown on a PC through UART1, so the activity can be monitored.

The system is simple, low-cost and secure, making it suitable for bank lockers and safes.